

GMxB

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Abstract. A variable annuity is a fund-linked insurance contract, including a package of financial options on the policy account value. Available guarantees are referred to as GMxB, where ‘x’ stands for the class of benefits involved. They may be first grouped into two main broad classes: Guaranteed Minimum Death Benefits (**GMDB**), and Guaranteed Minimum Living Benefits (**GMLB**). The second class can be further arranged into three subclasses: Guaranteed Minimum Accumulation Benefits (**GMAB**), Guaranteed Minimum Income Benefits (**GMIB**), and Guaranteed Minimum Withdrawal Benefits (**GMWB**). Variable annuities are generally issued with a *single premium* or *single recurrent premiums*. The total amount of premiums is also named the principal of the contract or the invested amount. Apart from some upfront costs, premiums are entirely invested into the reference funds chosen by the policyholder.

1. The Titanic Option: Valuation of the Guaranteed Minimum *Death Benefit* (GMdB) [1]

1.1. The death benefit definition

The net single premium for a life insurance contract paying $DB(t)$ at death is

$$\int_0^\infty DB(t)e^{-rt}f_x(t)dt,$$

where $F_x(t) = \Pr(\tau \leq t) = 1 - \exp(-\int_0^t \lambda(x+s)ds)$ is the cdf of the time of death τ , $f_x(t)$ its density (the instantaneous death probability at t), and the force of mortality $\lambda(x+t)$ is proportional to the entry age.

1.2. The discounted value of the insurance risk charge, $E_\lambda[F_\tau]$

We aim to find a fair insurance risk charge l that equates the present value of fees and benefits, assuming a constant risk charge. The insurance company holds a long position in the continuous fee stream, which survives until the policyholder dies or lapses.

$$\begin{aligned} dS_t &= (r_t - l_t)S_t dt + \sigma(S_t, t)S_t dB_t \\ F_\tau &= 1 - R_\tau^{-1}S_\tau + \int_0^\tau R_t^{-1}\sigma(t, S_t)S_t dB_t \\ E_\lambda[F_{\min[\tau, T]}] &= E_\lambda \left[\int_0^{\min[\tau, T]} l e^{-lt} dt \right] = 1 - \int_0^{\min[\tau, T]} e^{-lt} f_x(t) dt \\ &= 1 - \int_0^T e^{-lt} f_x(t) dt - P(\tau \geq T) e^{-lT} \end{aligned}$$

The integral represents the discounted value of the insurance fees paid, conditional upon death occurring before the policy termination time T , whilst the second component represents the event of survival to time T .

$$= 1 - E_\lambda[e^{-l\tau}] < 1 - e^{-lE_\lambda[\tau]}$$

Last inequality (with undefined policy termination) is a consequence of Jensen's inequality, which implies that fees are non linear concave function wrt expected death time (τ), ie $\partial\phi/\partial\tau \geq 0$, whereas $\partial^2\phi/\partial\tau^2 \leq 0$, i.e. evaluating expected discounted fees with the mean death time will overestimate fees.

1.3. Valuation of the GMdB Option, $E_{S,\lambda}[V^g]$

$$\begin{aligned} e^{-r\tau} \max(S_\tau, G_\tau) &= e^{-r\tau} (S_\tau + \max(G_\tau - S_\tau, 0)) = e^{-r\tau} S_\tau + E_{S,\lambda}[V^g] \\ &= \text{Account Value} + \text{GMdB Option Payoff} \end{aligned}$$

1.3.1 Valuation of Rising-Floor GMdB

Without a cap, the guaranteed rising-floor rate g cannot exceed the risk-free rate to avoid internal arbitrage. The GMdB put price is a mortality-weighted average of deterministic puts with dividend yield l , over the interval $[0, T]$:

$$E_{S,\lambda}[V^g] = \int_0^T D_r(0, t) \left(\int_0^\infty \max(G_t - x, 0) f_S(x) dx \right) f_x(t) dt = \int_0^T BSM(\dots) f_x(t) dt$$

The BSM_t put price is not globally convex/concave in time-to-maturity — it is hump-shaped: concave for short maturities, convex for long ones. Hence, by Jensen's inequality, for a given t the standard put with deterministic maturity is more expensive than a stochastic-maturity option whose expected maturity equals t when the function is concave; the relation reverses in the convex region.

1.3.2 Valuation of Look-Back GMdB

$$\begin{aligned} \text{GMdB Option Payoff at Death} &= \max_{0 \leq u \leq T} S_u - S_T \\ E[V^L] &= \int_0^T GSG(\dots) f_x(t) dt = \int_0^T GSG(\dots) \lambda(x+t) \exp\left(-\int_0^t \lambda(x+s) ds\right) dt \end{aligned}$$

1.4. Equating Fees and Benefits

What insurance fee will find the GMdb using the option pricing perspective? Locating a value of l — fixed risk-charge — that solves the identity, being aware that increasing the insurance charge will increase the present value of the insurance fees, but also increase the present value of the GMdB, recalling that a higher dividend yield increases the price of the put option in BSM, i.e. an equilibrium insurance charge may not exist,

$$\text{Economic Profit} = E_\lambda[F_\tau] - E_\lambda[V] = 0$$

2. The Real Option to lapse a Variable Annuity: Can Surrender Charges Complete the Market? [3]

Standard derivative-pricing models (Black–Scholes–Merton) assume the option premium is paid in full, upfront. Many insurance contracts with embedded options are instead paid in installments, through ongoing fees. This seemingly minor change has a profound consequence: it hands the policyholder a *real option to lapse*—the right to walk away precisely when the embedded guarantee is out-of-the-money, thereby avoiding payment for value the holder no longer wants. A holder who lapses and immediately repurchases at every new high can, in principle, ratchet a simple money-back guarantee (a vanilla put) up into a highest-value-achieved guarantee (a lookback put)—a far more expensive promise—without ever paying for the upgrade. The lapse option therefore breaks the hedge. Even when *both* the mortality and the financial markets are complete—so the insurer can fully diversify mortality risk

by selling enough policies and replicate any financial payoff—the holder’s freedom to lapse *de facto* renders the market incomplete: the contingent claim can no longer be hedged. Here is the central, and counterintuitive, result. Frictions—loads, commissions, surrender charges—are normally cast as the *source* of incompleteness: **for this claim, the friction is not what breaks the market—it is the only thing that completes it.** A properly calibrated contingent deferred surrender charge (DSC) is precisely the instrument that funds the guarantee and restores a self-financing hedge. Concretely, the model identifies the (π, κ) curve that exactly funds the guarantee when the charge is calibrated to the holder’s optimal lapse policy. The same curve delivers lower and upper fee bounds, π_L and π_U , outside which the product is unsustainable. Within this framework, lapsation is *rational* when it is immediately followed by a repurchase undertaken solely to reset the basis of the guarantee, and *irrational* when the surrender penalty exceeds the value of the new option.

2.1. Formulation of the Hedge Portfolio as American Contingent/Perpetual Claim

We assume a very large pool of insured fund owners—each of whom invests a relatively small amount in the protected mutual fund—so that a fraction $\lambda_t P(\tau \geq t) dt$ of the population dies between time t and $t + dt$. This is another way of stating the classical (and simplifying) assumption that mortality risk is completely diversifiable, and therefore not priced by the market. As a result, the outflow (or payout) due only to death between time t and $t + dt$ is precisely:

$$c_t dt = V_0 G(U_t) \lambda_t P(\tau \geq t) dt$$

where V_0 is the (very large) originally invested capital of all the fund owners (whilst U_t represents the fund units, i.e. $U_0 = 1$). The hedging portfolio will obey the following stochastic differential equation (where $c_t dt$ can be considered as a deterministic dividend):

$$dV_t + c_t dt = \alpha_t dS_t + \phi_t dR_t$$

The term $\pi_t dS_t + \beta_t dB_t$ in equation (10) is a self-financing portion, while the quantity $c_t dt$ represents the (consumption) outflow due to death. The value $Y_t - L_t$ is given by:

$$V_t \geq L = \begin{cases} V_0(1 - \kappa_t)P(\tau \geq T) U_t, & 0 \leq t \leq T, \text{ (Lapse)} \\ V_0 P(\tau \geq T) G(U_T), & t = T, \text{ (Maturity)} \end{cases} \quad (1)$$

The value of the portfolio (assets) must always exceed the liability. The portion $V_0(1 - \kappa_t)P(\tau \geq T) U_t$ covers the lapsation value of the fund, for the fraction $P(\tau \geq T)$ who are still alive, while $V_0 P(\tau \geq T) G(U_T)$ covers the maturity guarantee. The martingale representation of the self-financing hedging portfolio ($V_t + \nu R_t$, with $d\nu = (c_t/R_t)dt$ constructed due to dividends) is :

$$V_0 = M_0 = \tilde{E}_0 \left[\frac{1}{R_t} (V_t + \nu R_t) \right] = \tilde{E}_0 \left[\frac{V_t}{R_t} + \int_0^t \frac{c_s}{R_s} ds \right] = \tilde{E}_0 \left[\frac{V_t}{R_t} + \int_0^t \frac{V_0 \lambda_s P(\tau \geq T) G(U_s)}{R_s} ds \right], \quad (2)$$

for any $t \geq 0$, where $\tilde{E}_t[\cdot]$ denotes the risk-neutral conditional (on time- t) expectation. In words, the discounted value of the hedging portfolio plus the discounted sum of all payments made at death is a martingale. If, for example, the death rate and the continuous insurance charge are set to zero ($\pi_t = 0$, $\mu_t = 0$), while the deferred surrender charge is set at 100% (i.e., $\kappa_t = 1$), equation (13) with $t = T$ collapses to

$$V_0 = \tilde{E}_0 \left[\frac{\max(1, S_T)}{R_T} \right],$$

which is the Black–Scholes–Merton risk-neutral expectation for an at-the-money European put option plus a position in the underlying security.

2.2. The DSC/Fees optimal curve

If the insurance charge is exactly equal to π_U , the optimal lapsation level hits a value of one. This means that one should lapse the contract instantaneously after purchasing it, or in other words, the

contract should never be purchased. Stated differently, if $\pi > \pi_U$, the contract is not viable since no rational individual will ever hold it. We emphasize that all combinations of (π, κ) on the curve represent an appropriate charge for the embedded option. In other words, there is no unique price for the contingent claim. Either of these pairs—or any combination on the curve in between—will create enough money to construct a self-financing portfolio to hedge the option. Of course, the optimal lapsation policy on the part of the individual will depend on which one of these pairs is chosen. One can say that transaction costs (frictions) complete the market and allow for the existence of a self-financing strategy. The self-financing strategy does not result in a unique price, per se, but rather a menu or schedule of charges that can support the claim.

2.3. Hedging the optimal lapsation boundary

The effect of lapses and surrenders on variable annuities is ambiguous. Early termination reduces fee income and may leave the guarantee under-funded, effectively creating an incomplete market. Under rational lapsation, one can study the optimal time to lapse and its impact on guarantee valuation. In practice, re-establishing the cost basis through selling and repurchasing incurs a load proportional to the account value, declining over time. Assuming full rationality, the insurer might hedge by buying an up-and-out put option whose knockout barrier is the optimal lapsation boundary.

3. The Guaranteed Minimum Withdrawal Benefit (GMWB): A Quanto Asian Put in Disguise [2]

The GMWB guarantees a minimum annual withdrawal of G for as long as the initial invested capital has not been fully returned, regardless of the sub-account performance. We assume continuous withdrawals at the guaranteed rate, so the maturity of the guarantee is $T = w_0/G$.

3.1. The account value and the “ruin” integral

Let F_t be the price of one unit of the fee-paying sub-account (drift $\mu - \alpha$). We normalise $F_0 = 1$ so that the initial investment w_0 buys exactly w_0 units. The account value W_t (in dollars) satisfies

$$dW_t = (\mu - \alpha)W_t dt - G dt + \sigma W_t dB_t, \quad W_0 = w_0,$$

while $W_t > 0$. The dollar withdrawal $G dt$ is met by selling G/F_t fund units at the prevailing price F_t . The number of units remaining at time t is therefore

$$\frac{W_T}{F_T} = \frac{w_0}{F_0} - \int_0^T \frac{G}{F_s} ds = w_0 - G \int_0^T F_s^{-1} ds = w_0 - G \int_0^T e^{-(\mu - \alpha - 0.5\sigma^2)s - \sigma B_s} ds.$$

Multiplying back by F_t gives the dollar account value, with the absorbing ruin boundary:

$$W_T = F_T \max\left(0, w_0 - G \int_0^T F_s^{-1} ds\right).$$

The integral $I_T = \int_0^T F_s^{-1} ds$ acts as a **stochastic clock** that measures how many “time-zero unit equivalents” have been consumed by withdrawals. When the fund price is low, $1/F_s$ is large and the clock ticks faster; when the fund performs well, the clock slows down. Ruin occurs exactly when the clock reaches the fixed time budget w_0/G :

$$\xi_T = \Pr\left(\int_0^T \frac{1}{F_s} ds \geq \frac{w_0}{G}\right).$$

3.2. Decomposition into annuity and Quanto Asian Put

Under the risk-neutral measure, the GMWB is a bundle of a term-certain annuity and this exotic option, an **Asian Quanto** option on A which is the average of the reciprocal account value, as the

guarantee being path-dependent: it protects against the risk that the fund performs poorly *over the entire withdrawal period*, not just at the end.

$$\begin{aligned} \text{PV}_{\text{GMWB}} &= e^{-rT} G \int_0^T e^{rt} dt + e^{-rT} \mathbb{E}^Q[W_T] = \underbrace{\frac{G}{r}(1 - e^{-rT})}_{\text{term-certain annuity}} + e^{-rT} \mathbb{E}^Q[W_T] \\ &= \frac{G}{r}(1 - e^{-rT}) + w_0 e^{-rT} \mathbb{E}^Q\left[\frac{\max(1 - A, 0)}{Y_T}\right] = \frac{G}{r}(1 - e^{-rT}) + w_0 e^{-rT} \mathbb{E}^Q\left[\frac{\max(1 - \frac{1}{T} \int_0^T F_t^{-1} dt, 0)}{Y_T}\right]. \end{aligned}$$

where $Y_t = F_t^{-1}$ as the number of VA sub-account units that 1 dollar can buy,

3.3. Fair fee condition

The contract is fairly priced if the initial value equals the invested amount. Dividing by w_0 and setting $g = G/w_0$ gives the self-financing condition

$$e^{-r/g} \mathbb{E}^Q\left[\frac{\max(1 - A, 0)}{F_T}\right] + \frac{g}{r}(1 - e^{-r/g}) = 1.$$

For given market parameters (r, σ) and guarantee rate g , this equation determines the fair insurance fee α . The decomposition above forms the **core valuation representation** of the GMWB. Additional features—such as rational lapsation or surrender charges—complicate this baseline because they give the policyholder an American-style option to exit when the guarantee is worthless.

References

- [1] Moshe A. Milevsky and Steven E. Posner. The titanic option: Valuation of the guaranteed minimum death benefit in variable annuities and mutual funds. *Journal of Risk and Insurance*, 68(1):93–128, 2001.
- [2] Moshe A. Milevsky and Thomas S. Salisbury. Financial valuation of guaranteed minimum withdrawal benefits. *Insurance: Mathematics and Economics*, 38(1):21–38, 2006. ISSN 0167-6687. . URL <https://doi.org/10.1016/j.insmatheco.2005.06.012>.
- [3] Moshe A. Milevsky and Thomas S. Salisbury. The real option to lapse and the valuation of variable annuity guarantees. *Journal of Risk and Insurance*, 73(3):509–534, 2006.